HR ATTRITION ANALYTICS

Findings & Recommendations Report

IBM HR Employee Attrition Dataset | 1,470 Employee Records

Tool: PostgreSQL + Power BI

Executive Summary

This report presents a data-driven analysis of employee attrition across 1,470 records from the IBM HR dataset. Using PostgreSQL for SQL analysis and Power BI for visualization, six key business questions were investigated to identify the primary drivers of employee turnover.

The company currently experiences a 16.1% attrition rate, with 237 employees having left the organization. The analysis reveals that salary, overtime, department, and age are the most significant predictors of attrition — all of which are actionable by HR leadership.

Key Metrics at a Glance

Metric	Value	Insight
Total Employees	1,470	Full dataset analyzed
Employees Who Left	237	16.1% attrition rate
Avg Income — Attrited	$4,787 / month	30% below retained avg
Avg Income — Retained	$6,832 / month	Baseline comparison
Highest Attrition Dept.	Sales — 20.6%	2x higher than R&D
Lowest Satisfaction Role	Sales Representative	2.31 out of 4.0
Overtime Attrition Rate	30.5%	3x non-overtime rate
Highest Risk Age Group	Under 25	~38% attrition rate

Detailed Findings

Finding 1 — Overall Attrition Rate: 16.1%

Out of 1,470 employees, 237 have left the organization — representing a 16.1% attrition rate. This is significantly above the industry benchmark of 10–12% for most sectors, indicating a systemic retention problem that requires immediate attention.

Finding 2 — Sales Department Has Highest Attrition

Attrition by department reveals a clear hierarchy of risk:

- Sales: 20.6% attrition rate — highest across all departments
- Human Resources: 19.0% — second highest
- Research & Development: 13.8% — lowest, but still above industry benchmark

The Sales department attrition rate is nearly

Finding 3 — Salary is the Strongest Attrition Driver

The income gap between employees who left and those who stayed is significant:

- Employees who left earned an average of $4,787/month
- Employees who stayed earned an average of $6,832/month
- This represents a $2,045/month gap — a 30% income difference

Attrition by salary band further confirms this pattern:

Salary Band	Attrition Rate	Risk Level
Under $2,000/month	~53%	Critical
$2,000 – $5,000/month	~20%	High
$5,000 – $10,000/month	~11%	Medium
Above $10,000/month	~9%	Low

Finding 4 — Job Satisfaction Lowest Among Sales Representatives

Average job satisfaction scores (rated 1–4) by role reveal that front-line and technical roles consistently report lower satisfaction:

- Sales Representative: 2.31 / 4.0 — lowest satisfaction, 39.8% attrition
- Human Resources: 2.55 / 4.0
- Laboratory Technician: 2.68 / 4.0
- Manager: 3.22 / 4.0 — highest satisfaction, 5% attrition

There is a strong inverse correlation between satisfaction and attrition — roles with the lowest satisfaction scores consistently show the highest turnover rates.

Finding 5 — Overtime is a Critical Attrition Predictor

Overtime has the most dramatic impact on attrition of any single variable analyzed:

- Employees working overtime: 30.5% attrition rate
- Employees not working overtime: 10.4% attrition rate

Overtime workers are

Finding 6 — Under-25 Age Group Has Highest Attrition

Attrition by age group reveals that early-career employees are the most at-risk segment:

Age Group	Attrition Rate	Risk Level
Under 25	~38%	Critical
25 – 34	~20%	High
55+	~17%	Medium
45 – 54	~15%	Medium
35 – 44	~10%	Low

Young employees under 25 leave at nearly 4x the rate of the 35–44 age group, pointing to gaps in onboarding, mentorship, and early career development programs.

Recommendations

Recommendation 1 — Review Compensation for High-Risk Roles

Priority: High | Timeline: 0–3 months

- Conduct a market compensation benchmarking study for Sales Representative and Laboratory Technician roles
- Introduce performance-linked bonuses for front-line employees in the under-$2k salary band
- Set a minimum salary floor of $3,000/month for all full-time positions
- Expected impact: Could reduce under-$2k band attrition from 53% to below 30%

Recommendation 2 — Address Overtime Culture

Priority: High | Timeline: 1–2 months

- Cap mandatory overtime at 10 hours per week for all non-managerial staff
- Introduce overtime compensation or time-off-in-lieu policies
- Track and report overtime hours at the department level in monthly HR reviews
- Expected impact: Reducing overtime attrition to non-overtime levels could retain ~60 additional employees annually

Recommendation 3 — Launch Early Career Retention Program

Priority: Medium | Timeline: 2–4 months

- Introduce a structured 6-month onboarding and mentorship program for employees under 25
- Create a clear career progression roadmap for junior roles in Sales and R&D
- Conduct monthly check-ins with employees in their first year
- Expected impact: Reducing under-25 attrition from 38% to 20% would retain ~25 employees per cohort

Recommendation 4 — Conduct Stay Interviews in High-Risk Departments

Priority: Medium | Timeline: 1–3 months

- Implement quarterly stay interviews for all employees in Sales and HR departments
- Use structured questions around workload, satisfaction, compensation, and career growth
- Create a feedback loop where results are actioned within 30 days
- Track satisfaction scores monthly using the JobSatisfaction metric as a leading indicator

Recommendation 5 — Build a Predictive Attrition Early Warning System

Priority: Low | Timeline: 3–6 months

- Use the five highest-correlated variables (salary, overtime, age, satisfaction, tenure) to build a risk score for each employee
- Flag employees with a risk score above 70% for proactive HR intervention
- Review flagged employees quarterly with department managers

Conclusion

This analysis demonstrates that employee attrition at the company is driven by a concentrated set of addressable factors — low compensation, excessive overtime, poor early-career support, and role-level dissatisfaction. The five recommendations above, if implemented in order of priority, could reduce overall attrition from 16.1% to below 10% within 12 months.

The Power BI dashboard built alongside this report enables HR business partners to monitor these metrics in real time, filter by department and role, and track the impact of interventions over time.

Report prepared using IBM HR Employee Attrition Dataset (Kaggle) — 1,470 records, 33 variables. Analysis conducted in PostgreSQL with visualization in Power BI.